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6.3.3 Dot product: error results

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error analysis, we often like to do better than that. We often would like to know exactly what error can be attributed to what element in the vector, as opposed to lumping all of the elements together and talking about the norm of the vector. So for that reason, we introduce a new notation. Let's motivate that notation. If we now look at Δy more carefully by exposing its individual entries as well as the individual

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Reading assignment

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Read Unit 6.3.3 of the notes. [\[LINK\]](#)

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Homework 6.3.3.1

1/1 point (graded)
Let $k \geq 0$ and assume that $|\epsilon_1|, |\epsilon_2| \leq \epsilon_{\text{mach}}$, with $\epsilon_1 = 0$ if $k = 0$.
Show that

$$\begin{pmatrix} I + \Sigma^{(k)} & 0 \\ 0 & (1 + \epsilon_1) \end{pmatrix} (1 + \epsilon_2) = (I + \Sigma^{(k+1)}).$$

Hint: Reason the cases where $k = 0$ and $k = 1$ separately from the case where $k > 1$.

☒ done



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✓ Correct (1/1 point)

Homework 6.3.3.2

1/1 point (graded)
Prove Corollary 6.3.3.2. Part R1-B.

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Solution:

For a complete solution, see [the notes](#).

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